Total protein quantification by Bradford

What you're going to get / What you expect

The protein concentration (mg/mL) of each sample, needed to normalise activity into specific activity.

Materials list (per sample (96-well format))

- Protein assay dye reagent concentrate (Bradford).
- Bovine serum albumin (BSA) standard, range 0 to 0.5 mg/mL.
- 96-well plate; microplate reader (595 nm).

Steps

1. Build a calibration line with known BSA concentrations across the range 0 to 0.5 mg/mL.
2. Add 200 uL of Bradford reagent to 10 uL of enzyme solution.
3. Measure absorbance in triplicate at 595 nm.
4. Interpolate the sample readings on the BSA standard line to obtain the net protein concentration in mg/mL.
5. For concentrated stocks, use serial dilutions so the reading falls within the linear range of the standard.

Troubleshooting

- Reading above the standard range: dilute the sample serially until it falls inside the linear range.
- Always run in triplicate to reduce pipetting scatter.